Hyung Cho

AI Engineer | LLM Evaluation and Release Quality | Validation Engineer by Training

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SUMMARY

Validation engineer turned AI engineer. Led product validation at Zebra Technologies for 10 years: test plans, acceptance criteria, failure isolation, and final reports for 6 shipped hardware products. Since 2025, builds LLM applications and the evals that gate them on the Chat Completions, Responses, and Messages APIs.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Built a 60-question, five-category retrieval eval that scores vector, keyword, and hybrid search on recall@k, MRR, and nDCG@5 and posts the PR-versus-main difference on pull requests; hybrid search scores 0.950 recall@5.
- Designed an escalation eval whose LLM judges come from two providers, OpenAI and Anthropic, with a weekly 5% false-resolve ceiling and a judge-agreement check; it has flagged a breach four weeks running since the question set grew.
- Made model choice configurable per role across OpenAI, Azure OpenAI, and OpenAI-compatible endpoints, and migrated the eval answer generator to a newer model with the judges held fixed.
- Set up scheduled eval jobs that commit dated results and open an issue when a run fails or measures nothing, and added fail-closed guards after a benchmark published 0.000 for 42 runs while the README still showed 1.000.
- Built a framework-free tool-calling loop on the raw OpenAI SDK with streamed responses, a 5-iteration cap, and a sub-agent that reads long documents in its own isolated context.

Rep Today, native iOS workout app, pre-release

- Built a Cloudflare Worker proxy to OpenAI for a premium coaching chat, with server-side App Attest and subscription checks, capped output, and 462 proxy tests in CI.
- Grew the iOS unit suite from 829 to 1,687 tests, run on iOS 18 and iOS 26 in CI with 0 failures, and fixed platform defects CI caught, including a layout loop that hung the suite past the 45-minute job limit.

CreativeBridge, K-12 AI story-writing iOS app, internal TestFlight builds

- Took a TypeScript codebase from 961 type errors and 790 failing tests to zero in two weeks by fixing, skipping, or retiring tests, then made type-check, lint, and tests blocking in CI.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Fine-tuned open-source LLMs (LLaMA, Falcon) and built a chatbot on a commercial model, with an evaluation workflow.
- Built DeepLabV3+ and UNET segmentation models for cancer-cell histopathology and Sentinel-2 satellite imagery.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Led engineering, design, and product validation for 6 shipped hardware products, from test plans to final reports.
- Owned SDK validation across Windows, macOS, and Linux: compatibility, performance benchmarks, and reliability.
- Built C# test automation that replaced manual workflows and cut regression cycles by about 30%.
- Mentored 10 test technicians and set the lab's quality standards.

SKILLS

LLM APIs: OpenAI API, Anthropic API, Azure OpenAI, Tool calling, RAG, Prompt engineering
Evaluation: LLM evaluation, LLM-as-judge, RAGAS, Retrieval metrics, Regression testing, Performance testing
Engineering: Python, TypeScript, JavaScript, SQL, FastAPI, Pydantic, PostgreSQL, Cloudflare Workers, Docker
Quality and delivery: Test planning, Test automation, Failure analysis, GitHub Actions, CI/CD, XCTest, Claude Code

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo
FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses